China Strategies in the Automotive Industry

Seminar Paper

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1 Introduction

China is the world's largest automotive market and the global epicenter of the transition to electric mobility. In 2025, electric vehicles accounted for approximately 55% of new car sales in China, and Chinese-headquartered manufacturers supplied 60% of global electric vehicle sales that year (IEA, 2026). This rapid shift has exposed fundamental differences in how automotive firms are built and governed and has produced sharply divergent competitive outcomes across three distinct organizational models.

State-owned enterprises operating through mandated joint ventures with foreign partners, the dominant model of the reform era, are losing domestic market share at an accelerating rate. Privately built domestic champions, by contrast, have emerged as global market leaders, while acquisition-driven conglomerates have used cross-border M&A to rapidly close capability gaps with established Western brands. The literature suggests that these diverging trajectories are not coincidental: political governance constraints systematically impede strategic adaptation relative to private ownership structures (Shleifer & Vishny, 1994; Megginson & Netter, 2001).

Against this backdrop, this paper examines which governance and financing structure is most likely to create long-term shareholder value in a technology-intensive, high-growth market. Three cases are compared: SAIC Motor, representing the state-directed joint venture model; BYD, representing organic vertical integration; and Geely, representing M&A-driven private expansion. China's automotive industry offers a rare opportunity to evaluate these three models under identical market conditions, making it a particularly instructive case for broader questions of industrial policy and SOE governance.

SAIC Motor is China's largest state-owned automaker and the most prominent representative of the joint venture model. Majority-controlled by the Shanghai SASAC (63.71%) and listed on the Shanghai Stock Exchange, the firm has operated since the 1980s as the state-designated partner of Volkswagen and General Motors, with both ventures structured as 50/50 partnerships under China's quid pro quo industrial policy regime (Holmes et al., 2015; Bai et al., 2025). This arrangement produced stable rents and broad capability transfer for more than two decades but has come under acute pressure since 2024: net profit collapsed by 88%, JV sales fell by more than half, and the European Commission imposed a 35.3% countervailing duty on its EV exports (SAIC Motor, 2025; European Commission, 2024). BYD, founded in 1995, represents the vertically integrated private model. By controlling approximately 70 to 80% of its core components internally and building its manufacturing capabilities organically ahead of state industrial policy (Whitfield & Wuttke, 2026), it has become the global market leader in new energy vehicles (BYD Annual Report, 2025). Geely pursued a different path, acquiring Volvo Cars from Ford in 2010 for USD 1.8 billion (Balcet et al., 2012) to gain immediate access to advanced vehicle platforms and a global distribution network that would otherwise have taken decades to build organically (Luo & Tung, 2007). The investment was validated by the 2021 Volvo Cars IPO at approximately USD 18.5 billion.

The analysis is structured along four dimensions, namely governance, financing, capability building, and long-term viability, and draws on audited annual reports, primary regulatory sources, and peer-reviewed academic literature. The findings indicate that governance structure is the binding constraint across all dimensions: it determines which financing strategies are viable, which capabilities can be developed autonomously, and ultimately which competitive positions remain accessible under rapid technological change.

2 Governance

2.1 SAIC Motor: State Ownership and Political Governance

SAIC Motor Corporation Limited operates under a two-tier state ownership structure. Shanghai Automotive Industry (Group) Co., Ltd., itself wholly controlled by the Shanghai State-owned Assets Supervision and Administration Commission (Shanghai SASAC), holds 63.71% of SAIC Motor's listed shares (SAIC Motor, 2026). Combined with additional state-owned blockholders including China COSCO Shipping (5.91%) and Yuejin Motor Group (3.60%), the cumulative state share reaches approximately 73%. Although SAIC Motor is publicly listed on the Shanghai Stock Exchange (code 600104), control rights over board appointments, strategic approvals, and capital allocation remain with SASAC. This ownership pyramid is the canonical civil-law mechanism for retaining government control over nominally privatized firms (Bortolotti & Faccio, 2009).

The role of the Chinese Communist Party in SAIC's governance is explicitly documented in the company's audited annual report. SAIC's 2025 Annual Report states that the firm operates under the principle of "two consistencies": adherence to "the Party's leadership over SOEs" and adherence to "the establishment of a modern enterprise system as the direction of SOEs reform" (SAIC Motor, 2026). This dual mandate confirms that Party governance functions not as an informal influence but as a formal, constitutive element of the firm's governance structure.

Shleifer and Vishny's (1994) foundational model of politician-firm interaction provides the analytical framework needed to interpret this institutional arrangement. Their central insight is that the allocation of real resources within a firm depends not on the allocation of cash flow rights but on the allocation of control rights (Shleifer & Vishny, 1994). Privatization that transfers equity stakes without transferring control therefore produces little change in firm behavior. Shleifer and Vishny further demonstrate that politicians actively prefer managers with weak equity stakes, because well-incentivized managers are better able to resist political demands (Shleifer & Vishny, 1994). Applied to SAIC, this framework predicts that the SASAC-controlled ownership structure and the formally documented Party mandate will systematically redirect firm behavior away from shareholder value maximization toward political objectives.

Fan, Wong and Zhang (2007) offer the most directly applicable empirical evidence for the Chinese context. Examining 790 partially privatized Chinese IPO firms over the period 1993–2001, they find that 27% have politically connected CEOs, defined as current or former government officials (Fan et al., 2007). These firms underperform politically unconnected peers by approximately 18% in three-year post-IPO cumulative abnormal returns and exhibit substantially weaker accounting performance: sales growth lower by 21% and earnings growth lower by 24% (Fan et al., 2007). Board composition reinforces the pattern: politically connected firms appoint bureaucrats to 36% of directorships compared with 19% in unconnected firms, while the share of directors with professional backgrounds is lower by 13 percentage points (Fan et al., 2007). The authors interpret these findings as consistent with the "grabbing hand" hypothesis: bureaucratic management extracts resources from listed SOEs to satisfy political objectives at shareholders' expense (Fan et al., 2007). SAIC's governance structure places it squarely within the population for which these effects are documented, suggesting that the performance penalties associated with political connectivity are structurally embedded rather than incidental.

In 2025, SAIC formally restructured its governance to comply with China's revised Company Law: the Board of Supervisors was abolished, with its statutory functions transferred to a strengthened Audit Committee of the Board of Directors (SAIC Motor, 2026). While this reform represents formal modernization, it does not alter the underlying ownership structure or the documented Party leadership mandate. The persistence of the 63.71% controlling SASAC stake, combined with the explicit "two consistencies" principle, means that the structural conditions identified by Shleifer and Vishny (1994) and empirically confirmed by Fan et al. (2007) remain in place. As Megginson and Netter (2001) demonstrate in their comprehensive review of privatization evidence across 41 countries, mixed-ownership enterprises are not significantly more profitable than pure state-owned firms: full control transfer, not procedural modernization, is required to alter the underlying incentive structure.

2.2 BYD: Founder-led Private Governance

BYD is controlled by its founder Wang Chuanfu, who has served simultaneously as Chairman and CEO since the company's founding in 1995 and directly holds approximately 17% of shares (BYD Annual Report 2024). Together with co-founder Lu Xiangyang, who holds around 5.1% through Youngy Investment and serves as Non-Executive Director on the board, the founding team commands more than one third of voting rights under a one-share-one-vote structure (BYD Annual Report 2024). Other significant shareholders including Berkshire Hathaway and BlackRock with approximately 3.0% each have no operational decision-making authority and are not represented on the board. The company is privately organised; state subsidies amounted to RMB 10.4 billion in 2024 but confer no control rights (BYD Annual Report 2024). A party committee with strategic directive authority as structurally embedded in state-affiliated enterprises does not exist at the operational level. Unlike SAIC Motor, which historically built its market position on state-mandated joint ventures with Western OEMs, BYD emerged as an independent private manufacturer without JV dependency a structural freedom that shaped its governance model from the outset (Yuan & Brasó Broggi 2025).

This governance structure has direct consequences for decision-making speed. Wang Chuanfu explicitly identified a "fast decision-making mechanism" as one of BYD's three central competitive advantages alongside core technology and strategic direction (Wang Chuanfu, cited in He 2025). Strategic decisions the complete exit from combustion engine production in 2022, the early entry into semiconductor manufacturing through BYD Semiconductor, and the formal spin-off of the FinDreams Group in 2020 were implemented without consensus processes involving external shareholders or political authorities (BYD Annual Report 2024; He 2025). Zhang et al. (2025) provide empirical support for this mechanism: founder control in Chinese high-tech firms promotes breakthrough innovation because it shields against short-term capital market pressure and concentrates technical expertise with decision-making authority. Wang Chuanfu's background as a trained chemist corresponds to the ideal type of founder-technologist identified in the paper (Zhang et al. 2025).

A key advantage arises from this ownership structure: because Wang Chuanfu retains control and no broad shareholder base demands short-term profit distributions, BYD can consistently reinvest in long-term projects a discipline that would be difficult to sustain under dispersed ownership with quarterly earnings pressure. In 13 of 14 years between 2011 and 2024, BYD's R&D expenditures exceeded annual net profit (He 2025); in 2024 alone, RMB 54.2 billion was invested in research and development an increase of 35.7% year-on-year even though net profit in the same year amounted to RMB 40.3 billion (BYD Annual Report 2024). Xia (2008) shows, based on 229 Chinese listed companies, that investors place greater trust in founder-controlled firms than manager-controlled ones: the capital market treats founder control as a signal of reliable corporate governance and rewards it with a higher market valuation (Xia 2008). Xu & Wang (1999) complement this finding at the institutional level: in an analysis of Chinese listed companies, they show that ownership concentration correlates positively with profitability, while state ownership has a neutral to negative effect. BYD's founder-controlled, state-free ownership structure thus corresponds to the empirically superior governance type (Xu & Wang 1999).

2.3 Geely: M&A-Driven Private Governance

Geely's governance model represents a third distinct archetype in Chinese automotive development: a privately held conglomerate that deploys cross-border M&A as its primary capability-building mechanism. Zhejiang Geely Holding Group (ZGH) is wholly owned by founder and chairman Li Shufu, with ZGH-related vehicles collectively controlling approximately 76% of the publicly listed Geely Automobile Holdings (HK: 0175), including ZGH directly (39.94%) and Proper Glory Holding Inc. (26.2%), while Li Shufu retains sole private ownership of the unlisted ZGH apex entity. This concentrated founder-ownership structure, unencumbered by state committee approval processes or joint-venture partner consent, confers the strategic agility to execute rapid, large-scale foreign acquisitions.

Luo and Tung's (2007) springboard perspective provides the appropriate theoretical anchor: emerging-market firms use cross-border acquisitions of developed-market assets as springboards to overcome latecomer disadvantages in technology, brands, and management competence. ZGH has operationalised this logic across a sequenced portfolio of transactions: the USD 1.8 billion acquisition of Volvo Cars from Ford Motor Company in 2010, the acquisition of LEVC (London Electric Vehicle Company) in 2013, a 51% controlling stake in Lotus Cars in May 2017, and the formation of a 50:50 joint venture with Mercedes-Benz for the Smart brand in January 2020, with registered capital of RMB 5.4 billion.

The governance challenge of administering this multi-brand portfolio is resolved through differentiated integration rather than centralised control. Zheng, Noorderhaven and Du (2022, Journal of World Business, DOI: 10.1016/j.jwb.2021.101305) document how Geely managed the Volvo relationship through four sequential stages, distancing, balancing, building, and diversifying, driven by legitimacy pressures from both host-country stakeholders and acquired-firm employees. This arm's-length philosophy preserved Volvo's engineering culture while enabling capability transfer, most concretely through the jointly developed Compact Module Architecture (CMA) platform; analogous logic grants Lotus its British performance identity and the Smart JV formally co-equal board representation.

By contrast, SAIC Motor's state-directed governance constrains strategic autonomy through its SOE ownership structure, while BYD's founder-led model pursues capability through a vertically integrated model including battery and semiconductor production. Geely's approach trades the cost efficiency of vertical ownership for the speed of strategic asset acquisition and the international legitimacy conferred by established Western brands.

3 Financing

3.1 SAIC Motor: The JV Capital Model and Its Structural Limits

SAIC's historical financing model was capital-light by design. China's quid pro quo policy required foreign automakers to form 50/50 joint ventures with domestic partners as a condition of market access, meaning Volkswagen and General Motors each bore half of the investment costs and operational risk in their respective ventures with SAIC (Holmes et al., 2015). For more than two decades, this arrangement provided SAIC with stable dividend streams without requiring the firm to bear the full costs of vehicle development, manufacturing investment, or distribution infrastructure, an arrangement that systematically favored the domestic partner.

The structural weakness of this model became apparent in 2024. SAIC-GM's annual vehicle sales collapsed from approximately one million units in 2023 to 435,007 units in 2024, a decline of 56.54%, prompting a substantial asset impairment charge of which SAIC was required to absorb 50% (SAIC Motor, 2024; SAIC Motor, 2025). The financial impact was severe: consolidated revenue declined by 15.73% to RMB 627.59 billion, reported net profit fell by 88.19% to RMB 1.666 billion, and net profit excluding non-recurring items turned negative at RMB −5.41 billion (SAIC Motor, 2025). The weighted average return on equity dropped to 0.58%, down from 4.98% in 2023 and 5.84% in 2022 (SAIC Motor, 2024; SAIC Motor, 2025). What had been a capital-light arrangement thus became a liability: SAIC was contractually obligated to absorb half of JV losses, yet lacked the operational control needed to initiate restructuring of the foreign-led joint ventures.

SAIC's reported financial performance recovered substantially in 2025: net profit attributable to shareholders rose to RMB 10.11 billion, an apparent increase of 506% year-on-year (SAIC Motor, 2026). This headline figure, however, largely reflects the absence of the one-time 2024 impairment rather than an underlying operational improvement. The more informative measure, net profit excluding non-recurring items, improved from RMB −5.41 billion in 2024 to RMB 7.42 billion in 2025, representing a genuine recovery from operating losses, but remains below the comparable 2023 level of RMB 10.04 billion (SAIC Motor, 2024; SAIC Motor, 2026). Weighted average ROE recovered to 3.43%, still significantly below SAIC's pre-crisis levels (SAIC Motor, 2026). The 2025 figures are best interpreted as stabilization rather than structural recovery: the operating losses of 2024 have been arrested, but the underlying profitability of the JV-centric business model remains substantially impaired.

The European Commission's 2024 anti-subsidy investigation provides an independent and precisely quantified measure of state subsidy intensity. Following a twelve-month investigation, the Commission imposed a definitive countervailing duty of 35.3% on battery electric vehicle imports from SAIC Group, more than double the rate applied to BYD (17.0%) and substantially above Geely (18.8%) (European Commission, 2024). The 35.3% rate, the maximum applied to non-cooperating producers, partly reflects SAIC's refusal to provide requested data, though the Commission's residual calculation is itself based on the highest subsidy rates found among cooperating producers. While SAIC has contested these findings, the differential between SAIC and its private-sector peers nonetheless serves as a regulatory indicator that SAIC's competitive position depends more heavily on state support than those of BYD or Geely. SAIC's ROE, 5.84% in 2022, falling to 0.58% in 2024 and partially recovering to 3.43% in 2025, is consistent with Megginson and Netter's (2001) finding that mixed-ownership enterprises do not achieve the efficiency gains of fully privatized firms.

3.2 BYD: Vertical Integration Financing

BYD's financing model differs fundamentally from debt-intensive growth strategies: the company funds its expansion primarily from ongoing operations. In fiscal year 2024, BYD generated operating cash flow of RMB 133.5 billion on revenues of RMB 777.1 billion (+29% year-on-year) and net profit of RMB 40.25 billion (+34%) (BYD Annual Report 2024). Cash reserves stood at RMB 154.9 billion at year-end, while interest-bearing liabilities declined to RMB 28.58 billion — representing only 4.9% of total liabilities (BYD Annual Report 2024). Internal financing is supplemented by state subsidies, which amounted to RMB 10.41 billion in 2024, equivalent to approximately 26% of annual net profit (BYD Annual Report 2024).

The central advantage of this model lies in independence from capital markets. Since BYD does not need to finance its investments through external debt or equity issuances, there is no pressure to prioritise short-term profitable projects over long-term research initiatives. In 2024 alone, R&D expenditures of RMB 54.2 billion exceeded annual net profit of RMB 40.25 billion — a structural reinvestment pattern that would be difficult to sustain under dispersed ownership with quarterly earnings pressure (BYD Annual Report 2024). A further structural strength is margin resilience: gross margin rose to a record 19.4% in 2024 — despite intense price competition in the Chinese automotive market (BYD Annual Report 2024). Wu (2025) confirms this relationship empirically: an analysis of BYD's financial statements shows that vertical integration directly reduces costs, improves margins, and increases research capacity. Vertical integration reduces dependence on external suppliers, thereby protecting the cash flow base from which investments are funded. The low leverage ratio additionally confers financial resilience, as no significant interest payments weigh on liquidity.

However, the model has structural limits. First, growth speed is constrained by internally generated liquidity. In the early growth phase, BYD depended on external capital: in 2008, Berkshire Hathaway invested USD 230 million for what was then approximately a 10% stake, providing BYD with access to international capital and reputation (BYD Annual Report 2024). Second, subsidy dependence carries political risk: state subsidies currently amount to approximately one quarter of net profit and are tied to the continuation of Chinese industrial policy. A reduction or elimination of these funds would immediately constrain investment capacity. This risk should be qualified, however: Jiang & Xu (2023) show, based on 196 Chinese NEV companies, that state subsidies primarily increase the quantity of innovations rather than their quality — BYD's structural innovation depth, sustained by vertical integration and proprietary R&D capabilities, would remain largely intact in the event of a subsidy reduction. Third, a cash-flow-based model requires stable operational profitability. Given ongoing price pressure in the Chinese market, signs of strain are already visible: in the third quarter of 2024, quarterly profit declined 33% year-on-year (CNBC 2025). Should margin pressure become structurally entrenched, the capacity for self-financing would come under strain as well.

3.3 Geely: Acquisition Financing

The Volvo acquisition of 2010 represents the paradigm case of leveraged, state-assisted cross-border acquisition financing in Chinese automotive history. Balcet, Wang and Richet (2012, IJATM) document the deal structure: ZGH paid USD 1.8 billion for 100% of Volvo Cars, including a USD 200 million seller note from Ford, with the balance in cash, while simultaneously mobilising a further USD 900 million in operational reserves, bringing total capital raised to approximately USD 2.7 billion. The financing coalition comprised loans from the Bank of China, China Construction Bank, and Export-Import Bank of China (~USD 2.1 billion in bank credit), equity from the listed Geely Auto arm, contributions from the city of Gothenburg, and approximately USD 932 million (~50% of total) from Chinese central government sources. The consequences for ZGH's balance sheet were severe: group debt surged from RMB 16.05 billion (2009) to RMB 71.07 billion (2010), pushing the debt-to-capital ratio to 73.47%, markedly above the industry high of approximately 60%, while ZGH simultaneously absorbed Volvo's pre-existing USD 3.5 billion in obligations.

The strategic logic was speed. Organic development of equivalent brand, platform, and market assets would have required decades; the acquisition compressed that timeline to a single transaction. The thesis was vindicated by Volvo Cars' IPO on Nasdaq Stockholm in October 2021, which valued the company at approximately USD 18.5 billion, more than ten times the 2010 acquisition price, while ZGH retained a ~75–80% ownership stake, realising substantial unrealised gains.

Geely's subsequent portfolio, however, carries material refinancing risk. Polestar (Nasdaq: PSNY) recorded a net loss of USD 1.17 billion in FY2023, received a going-concern qualification from auditor Deloitte, and held only USD 103.1 million in cash at end-2024; analysts estimate a further USD 2 billion in capital requirements through 2028, with ZGH committed as group-level guarantor. Lotus Technology has accumulated a deficit of approximately USD 3.2 billion as of FY2025, with revenue falling 44% year-on-year in FY2025.

The contrast to peer financing models is instructive. BYD's vertically integrated model builds capability organically, avoiding acquisition debt but constraining the speed and geographic breadth of expansion. SAIC Motor's joint venture model shares capital requirements with foreign partners, limiting balance-sheet exposure while also constraining strategic independence. Geely's leverage-for-speed approach carries the highest financing risk of the three models, with group-level debt-servicing capacity remaining structurally dependent on Volvo's sustained profitability.

4 Capability Building

4.1 SAIC Motor: Technology Transfer and the Capability Paradox

To understand how SAIC developed its manufacturing capabilities, China's quid pro quo industrial policy framework must be considered. The 1994 Automotive Industry Development Policy required foreign automakers seeking market access to form joint ventures with domestic partners under a 50% foreign equity cap, effectively exchanging market access for technology transfer (Bai et al., 2025; Holmes et al., 2015). Holmes, McGrattan and Prescott (2015) conceptualize this arrangement as an implicit tax on technology capital: knowing that any technology transferred to the joint venture effectively becomes Chinese property within the country, foreign firms had reason to withhold their most advanced technologies (Holmes et al., 2015). As the state-designated domestic partner of both Volkswagen and General Motors, SAIC operated at the center of this regime and emerged as one of its primary beneficiaries (Bai et al., 2025).

Bai, Barwick, Cao and Li (2025) provide the most direct causal evidence on the policy's effects. Drawing on within-product variation across multiple quality dimensions in J.D. Power's Initial Quality Study, covering nearly all car models produced in China between 2001 and 2014, the authors estimate that ownership affiliation under the quid pro quo regime contributed 8.3% of the total quality improvement in affiliated domestic models relative to non-affiliated counterparts, with the quality gap narrowing from 65% higher malfunction rates in 2001 to 33% by 2014. The primary transmission channels were worker mobility and overlapping supplier networks; formal patent transfer played a negligible role. SAIC Volkswagen and SAIC General Motors are both explicitly identified within the paper's ownership network, and SAIC's introduction of the Roewe brand in 2006 reflects the type of capability accumulation the JV regime enabled (Bai et al., 2025).

Howell (2018) documents a countervailing effect. Exploiting China's 2009 Phase 2 fuel economy standards as a quasi-experiment, she finds that JV-linked firms responded to the binding regulatory constraint by reducing rather than upgrading product quality: torque fell by 11%, prices by 15%, and vehicle weight by 65 kg relative to comparable firms without JVs. The mechanism is a cannibalization dynamic: JV rents reduce the incentive to invest in autonomous innovation, since upgrading own-brand technology would compete with the profitable foreign-brand income stream. Former Minister of Machinery He Guangyan described JV rents as "like opium" for Chinese firms, addictive in the short term and detrimental to independent capability development (cited in Howell, 2018, p. 1448).

Taken together, Bai et al. (2025) and Howell (2018) describe a capability-building paradox. The JV model transferred production quality but simultaneously suppressed the autonomous innovation required for next-generation capabilities. By 2014 SAIC produced substantially improved vehicles under the Roewe and MG marques, but had not developed the proprietary battery, power electronics, or software platforms that would later define competitive advantage in the EV era. Inkpen and Beamish (1997) argue that joint ventures become unstable once asymmetric learning erodes the knowledge-dependency that made the partnership mutually beneficial, a process already underway by 2014 and fully visible by 2025, when SAIC-GM operated at 37% of designed capacity and Volkswagen's dedicated MEB plant at 14% (SAIC Motor, 2026). SAIC's 2025 Annual Report documents the firm's response: NEV sales of 1.643 million units (+33.14%), the industry's first mass deployment of semi-solid battery technology, and the launch of the SAIC SHANGJIE brand with Huawei (SAIC Motor, 2026). These are substantive steps, but they entered commercial deployment well after BYD's equivalent investments had already begun generating sustained market position.

4.2 BYD: Internal Build

BYD's capability building follows an evolutionary internal logic that differs fundamentally from JV-based technology transfer or acquisition-driven capability acquisition. The company did not enter battery technology as an automaker — rather the reverse: founded in 1995 as a manufacturer of mobile phone and consumer electronics batteries, BYD already possessed deep knowledge in cell chemistry before entering vehicle production in 2003. Whitfield & Wuttke (2026) document that BYD began investing in automotive batteries from the mid-2000s — not in response to state industrial policy, but ahead of it, thereby actively shaping the policy agenda. Wang, Zhao & Ruet (2022) describe BYD's approach as "Specialised Vertical Integration": a selective integration concentrated on strategically complementary value chain stages, which deepens core competencies rather than diffusing them.

This model generates advantages that are structurally difficult to replicate. Teece, Pisano & Shuen (1997) show that sustainable competitive advantage arises from path-dependently accumulated tacit knowledge — knowledge embedded in organisational routines that cannot be transferred through licensing or talent recruitment. BYD's decades of internal development produce precisely this kind of non-imitability. Moreover, each mastered competency opens access to the next: Whitfield & Wuttke (2026) document how BYD leveraged existing capabilities from the consumer battery sector to build automotive-specific knowledge incrementally — from cell chemistry through battery packs and power electronics to vehicle software. Through this early positioning, BYD did not merely remain a technology follower but became a market architect whose development trajectory state subsidy programmes subsequently adopted (Whitfield & Wuttke 2026).

The model nevertheless has structural limits. First, internal capability building is time-intensive: while a competitor can acquire a new capability through acquisition within months, organic learning requires years or decades — a disadvantage during technological disruptions with compressed time windows. Second, the model is capital-intensive: investments occur long before their commercial exploitation, increasing dependence on stable cash flow sources. Third, path dependence carries the risk of path rigidity: a paradigm shift in technology would partially devalue BYD's accumulated knowledge capital, as any redirection would require fundamental reorganisation (Teece, Pisano & Shuen 1997). Finally, simultaneous mastery of highly heterogeneous competency domains places significant demands on management: integrating diverse technological domains within a single firm generates organisational complexity that becomes harder to coordinate as the company grows.

4.3 Geely: Acquisition-based Capability Building

The 2010 Volvo acquisition gave ZGH instantaneous access to a cluster of strategic assets that would have taken Geely comparable decades to develop organically: Volvo's safety engineering expertise, three cutting-edge vehicle platforms, accumulated brand equity, and a global distribution network of over 2,400 dealer outlets. In the academic literature this is characterised as strategic asset-seeking FDI, Geely did not acquire Volvo for Swedish market access but for the engineering and institutional capabilities embedded in the brand, capabilities that then served as the basis for upgrading Geely's own products and processes.

The mechanism of absorption is documented in the four-stage integration model proposed by Zheng, Noorderhaven and Du (2022, Journal of World Business, DOI: 10.1016/j.jwb.2021.101305): distancing, balancing, building, and diversifying. The flagship output of the building stage is the jointly developed Compact Module Architecture (CMA) platform, deployed across Volvo's 40-series models, the Lynk & Co brand, and Geely-branded vehicles, concrete evidence that the transfer evolved from one-way absorption into bilateral co-development. That the deal catalysed rather than substituted for internal investment is confirmed by a 600% increase in Geely's R&D expenditure within seven years of acquisition.

Konda, Slepnikov and Jin (2022, Asian Journal of Technology Innovation, 31(3), DOI: 10.1080/19761597.2022.2133777) supply the quantitative foundation: longitudinal patent portfolio analysis reveals measurable convergence in the innovation outputs of Geely and Volvo over the post-acquisition period, distinguishing Geely-Volvo as a rare case of genuine co-creation among emerging-market acquirers rather than a one-directional extraction of developed-market knowledge.

The contrast with BYD's capability-building model is structurally significant. BYD's vertically integrated model, including battery and semiconductor production, follows a slower organic trajectory, but confers full proprietary control with no integration risk. Geely's acquisition route compressed the capability timeline decisively, yet its success remained contingent on sustaining the arm's-length governance conditions that protected Volvo's engineering culture; any deterioration of that relationship would have undermined the very assets the acquisition was designed to absorb.

5 Long-term Viability

5.1 SAIC Motor: Joint Venture Erosion and the EV Transition

The structural decline of SAIC's joint venture model is most clearly reflected in capacity utilization figures. In 2025, SAIC General Motors utilized only 37% of its designed production capacity, manufacturing 534,000 vehicles against a maximum of 1.452 million (SAIC Motor, 2026). SAIC Volkswagen reached 55% utilization with 1.058 million units produced out of a possible 1.920 million. Particularly striking is the performance of Volkswagen's dedicated MEB electric vehicle plant: despite a designed capacity of 240,000 units, actual production amounted to only 34,464 vehicles, a utilization rate of 14% (SAIC Motor, 2026). These numbers do not reflect short-term production disruptions but a sustained loss of demand for the joint ventures' ICE-focused model range. In contrast, SAIC's own-brand operations performed considerably better, with the Passenger Vehicle Branch at 76% and SAIC GM Wuling at 95% utilization. This divergence suggests that the issue is not with SAIC's production capacity as such, but specifically with the market positioning of its foreign-brand joint venture portfolio.

This decline in JV competitiveness coincides with a fundamental transformation of the Chinese automotive market. According to the International Energy Agency, electric vehicles accounted for approximately 55% of new car sales in China in 2025, with monthly EV share exceeding 50% in eleven out of twelve months (IEA, 2026). Chinese automakers supplied 60% of global EV sales that year, and 70% of battery electric cars sold domestically were already priced below the average conventional vehicle (IEA, 2026). The IEA projects that EVs will represent more than 90% of new car sales in China by 2035 (IEA, 2026). This transition is therefore not a distant prospect but a structural shift that is already well underway, and one that places SAIC's ICE-heavy JV portfolio at a significant competitive disadvantage.

At the same time, SAIC's most obvious alternative growth strategy, expanding EV exports to Europe, faces significant external constraints. Following its 2024 anti-subsidy investigation, the European Commission imposed a countervailing duty of 35.3% on SAIC's electric vehicle exports (European Commission, 2024). This rate is more than double the 17.0% applied to BYD and substantially higher than the 18.8% levied on Geely. As a result, SAIC faces simultaneous pressure on its two most important revenue channels: falling JV sales in the domestic market and restricted export access in Europe.

In response to these pressures, SAIC has outlined a strategic shift it describes as the "Joint Venture 2.0" model in its 2025 Annual Report (SAIC Motor, 2026). The approach involves repurposing underutilized JV manufacturing capacity for own-brand electric vehicle production, with the SAIC SHANGJIE brand, developed in cooperation with Huawei, reaching market in September 2025. Whether this transition can compensate for the structural decline in JV profitability is the central empirical question for assessing SAIC's long-term viability. The governance structure documented in Section 2.1 provides grounds for caution: with SASAC holding a 63.71% controlling stake and strategic decisions subject to the formal Party leadership mandate, SAIC's capacity to execute the rapid capital reallocation a successful transition would require remains structurally constrained.

5.2 BYD

BYD has structurally completed the transition to electric mobility. Following the complete exit from combustion engine production in 2022, no ICE segment burdens capital allocation any further. For fiscal year 2025, the Annual Report shows more than one million exported vehicles for the first time, an increase of 140% year-on-year — with a presence in 119 countries, structurally reducing BYD's dependence on the Chinese domestic market (BYD Annual Report 2025). For the fourth consecutive year, BYD was the global market leader in the NEV segment. Allen et al. (2024) show that externally listed Chinese private companies — such as BYD with its primary listing on the Hong Kong Stock Exchange — achieve long-term capital market performance at global market level, while state-dominated A-share companies structurally underperform. The capital increase of USD 5.6 billion in March 2025, involving institutional long-term investors and sovereign wealth funds, signals sustained investor demand (BYD Annual Report 2025).

Structural risks nonetheless persist. The ongoing price war in the Chinese market leaves visible marks: gross margin fell from 19% (2024) to 18% (2025), and net profit declined by approximately 19% to RMB 32.6 billion — indicating that vertical integration can dampen but not fully absorb systemic price pressure (BYD Annual Report 2025). In parallel, the leverage ratio shifted from a net cash surplus to a net debt position as a result of capital-intensive global factory expansion, partially constraining the self-financing advantage described in Chapter 3. The complete exit of Berkshire Hathaway from its BYD stake by 2025 also removed a longstanding reputational anchor (CNBC 2025). For the export strategy, the EU countervailing duty of 17% — with a combined tariff of 27% including the standard import duty — remains a structural obstacle in the most profitable target market (EU Implementing Regulation 2024/2754).

5.3 Geely: Long-term Viability

Geely's FY2024 core performance is strong. Geely Automobile Holdings reported revenue of USD 33.4 billion (+32% year-on-year) and net profit of RMB 16.8 billion (+240%), while Volvo Cars, the anchor acquisition, delivered SEK 400.2 billion in revenue at a 6.8% core EBIT margin, its second consecutive record year. Market sentiment reflects this: Geely Auto shares rose approximately 39% in early 2026, outpacing BYD's concurrent +12%.

Portfolio complexity is the countervailing structural risk. Polestar (Nasdaq: PSNY) recorded a net loss of USD 1.17 billion in FY2023, received a going-concern qualification from auditor Deloitte, and held only USD 103.1 million in cash at end-2024, a 75% decline year-on-year, with ZGH committed as guarantor for bank facilities and analysts estimating a further USD 2 billion in capital requirements through 2028. Lotus Technology added a net loss of approximately USD 1.075 billion in FY2024, an accumulated deficit of ~USD 3.2 billion as of FY2025, and FY2025 revenue contracting 44% year-on-year. The Smart JV with Mercedes-Benz sold fewer than 2,000 units per month in China in the second half of 2023.

The EV transition introduces a strategic dependency structurally absent from BYD's model: Geely's electrification relies on platforms developed through Volvo Cars and the Smart JV, Volvo Cars a separately listed entity, the Smart JV a privately held 50:50 arrangement with Mercedes-Benz, neither of which ZGH controls unilaterally, rather than on vertically integrated battery production and EV platforms developed in-house.

The overall assessment is asymmetric. A high-performing core in Geely Auto and Volvo Cars sustains the group, but the surrounding portfolio of loss-making subsidiaries creates contingent liabilities that, if crystallised simultaneously, would materially strain ZGH's leveraged balance sheet and test the resilience of the acquisition-led model that generated the core strength in the first place.

6 Generalizable Lessons

The comparison of SAIC Motor, BYD, and Geely across four analytical dimensions yields three lessons that extend beyond the Chinese automotive context. Each addresses a strategic choice that firms in technology-intensive, high-growth markets must resolve: how to acquire capabilities, how to make decisions, and how to finance expansion.

6.1 The Mode of Capability Acquisition Determines Durability Across Technology Cycles

The three cases represent three structurally distinct approaches to capability building: mandatory technology transfer through state-directed joint ventures (SAIC), organic internal development (BYD), and strategic asset acquisition through cross-border M&A (Geely). Each approach produces capabilities, but with fundamentally different durability profiles under technological discontinuity.

SAIC's joint venture regime generated measurable spillovers, Bai et al. (2025) document an 8.3% quality improvement in affiliated domestic models relative to non-affiliated counterparts over the 2001–2014 period but simultaneously suppressed autonomous innovation through the rent cannibalization mechanism identified by Howell (2018). Once the knowledge gap between SAIC and its JV partners narrowed, the structural logic of the partnership eroded (Inkpen & Beamish, 1997), leaving a firm with improved production quality but without the proprietary battery, software, or power electronics platforms required for EV competition. BYD's organic model avoided this trap: path-dependent accumulation of tacit knowledge in cell chemistry, power electronics, and vehicle software (Teece et al., 1997) produced capabilities that are proprietary and structurally non-imitatable, built not in response to state industrial policy but ahead of it, thereby actively shaping the policy agenda (Whitfield & Wuttke, 2026). Geely's acquisition of Volvo compressed the capability timeline through strategic asset-seeking FDI (Luo & Tung, 2007) and, critically, avoided the cannibalization problem because the transfer was voluntary and contractually governed rather than state-mandated; arm's-length integration (Zheng et al., 2022) preserved Volvo's engineering culture while enabling bilateral co-development, concretely demonstrated by the jointly developed CMA platform and the measurable convergence in innovation outputs documented by Konda et al. (2022).

The generalizable lesson is that capability acquisition through mandatory technology transfer creates short-run quality gains but structurally displaces the autonomous innovation needed for next-cycle competition. Voluntary acquisition and organic development avoid this displacement at different costs, speed versus depth versus independence, and the appropriate choice depends on whether the primary competitive constraint is time or technological sovereignty.

6.2 Governance Structure is a First-Order Determinant of Strategic Adaptability

In markets undergoing rapid technological transition, the speed at which a firm can reallocate capital and reorient its product portfolio is itself a competitive advantage. The three cases demonstrate that this speed is not primarily a function of resources but of governance structure, specifically, the distribution of control rights among principals with different objective functions.

SAIC's governance architecture distributes control across SASAC (63.71% ownership), the formally codified Party leadership mandate, and 50/50 JV boards with Volkswagen and General Motors that require bilateral consensus for strategic decisions. Shleifer and Vishny (1994) demonstrate that political principals maximize employment and political stability rather than firm value; Fan et al. (2007) confirm empirically that politically connected Chinese firms underperform unconnected peers by approximately 18% in post-IPO cumulative abnormal returns and exhibit substantially weaker earnings and sales growth. SAIC's constrained response to the EV transition, the "Joint Venture 2.0" strategy emerging only in 2025, more than five years after BYD's competitive position was already established, is consistent with this prediction. BYD, by contrast, operates under consolidated founder control: Wang Chuanfu holds sufficient equity and board authority to implement decisions without external consensus, identifying "fast decision-making" as one of three central competitive advantages alongside core technology and strategic direction (cited in He, 2025). Zhang et al. (2025) provide empirical support for this mechanism, demonstrating that founder control in Chinese high-tech firms promotes breakthrough innovation by shielding decision-making from short-term capital market pressure and concentrating technical expertise with strategic authority. This structural freedom enabled the complete exit from ICE production in 2022, a decision that would have required multi-stakeholder approval under either an SOE or a JV governance structure. Geely occupies an intermediate position: Li Shufu's private ownership of the apex holding company confers acquisition agility, but the resulting multi-brand portfolio introduces coordination costs and governance complexity, as visible in the divergent performance trajectories across ZGH's subsidiaries and the contested strategic direction of loss-making entities.

The lesson generalizes directly: governance structures that concentrate control rights with commercially-oriented principals adapt faster to technological discontinuities than those that distribute them across political principals, joint-venture partners, or diversified portfolios.

When the rate of technological change exceeds the decision cycle of a firm's governance structure, the governance arrangement itself becomes a binding competitive constraint.

6.3 Capital Structure Determines Resilience When Markets Contract

The three financing models differ not only in their cost of capital but in their risk distribution under adverse market conditions, and this distinction became decisive in the 2024–2025 period.

SAIC's capital-light JV co-investment model operated efficiently in a growing market: Volkswagen and General Motors bore half of plant investment and operational risk, providing SAIC with stable dividend income without commensurate capital exposure. The structural flaw became visible only under market stress: when SAIC-GM's annual sales declined by 56.54% in 2024, SAIC was contractually obligated to absorb 50% of the resulting asset impairment without holding the operational control required to restructure the joint venture. Net profit attributable to shareholders fell 88.19% to RMB 1.67 billion; weighted average ROE collapsed to 0.58% (SAIC Motor, 2025). The capital-light arrangement had transformed into a capital trap. BYD's self-financing model, funded from operating cash flows with gross margins of 19.4% in 2024 (BYD Annual Report, 2024) proved structurally resilient: no JV partner losses to absorb, no external debt service to maintain, and no renegotiation of control rights required. Its vulnerability is different in kind: sustained price competition that compresses margins reduces the internal cash generation on which the entire investment programme depends, as visible in the gross margin decline from 19% to 18% and net profit contraction of approximately 19% in 2025 (BYD Annual Report, 2025). Geely's leveraged acquisition model generated the highest strategic return, the 2021 Volvo IPO valued at approximately USD 18.5 billion against a 2010 acquisition price of USD 1.8 billion (Balcet et al., 2012) but created contingent liabilities through loss-making portfolio entities whose group-level guarantees could, if simultaneously crystallised, strain ZGH's debt-servicing capacity. Megginson and Netter (2001) establish the comparative benchmark: mixed-ownership enterprises systematically underperform fully private firms in profitability and efficiency, a pattern consistent with SAIC's ROE trajectory across the 2022–2025 period.

The generalizable lesson is that capital-light models which share risk with partners capture upside efficiently in growing markets but expose the firm to structural liability in downturns precisely when operational control over the risk source is absent. Self-financing models trade growth speed for resilience; leveraged acquisition models trade balance-sheet safety for strategic speed, with long-term viability contingent on the acquired assets generating sufficient cash flow to service acquisition debt. In high-velocity markets, capital structure is therefore not a static optimization problem but a dynamic bet on the firm's ability to maintain operational control over the primary sources of its financial exposure.

7 Conclusion

To-Dos:

- ☐ Vergleichskapitel jeweils (ggf. Tabellen)
- ☐ Gedankenstriche minimieren (sieht zu KI aus)
- ☐ Conclusion
- ☐ Texte anpassen zu einem einheitlichen flüssigen Paper
- ☐ Literaturverzeichnis
- ☐ Authorenzuordnung
- ☐ KI-Annex
- ☐ Ehrenwörtliche Erklärung
- ☐ Präsentation Slides
- ☐ Alles gegenprüfen und Korrekturlesen
- ☐ Zitationen checken (vor allem Annual Reports)

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KI Annex

Shared git repository with shared wikis

Source rating

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